Advanced Linear Algebra: Foundations to Frontiers

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2.2.3 Orthonormal vectors and matrices ¶

02.2.3 Orthonormal vectors and matrices

fit width

A lot of the formulae in the last unit become simpler if the length of the vector equals one: If $\|u\|_2 = 1$ then

- the component of v in the direction of u equals

$$\frac{u^H v}{u^H u} u = u^H v u.$$

- the matrix that projects a vector onto the vector u is given by

$$\frac{u u^H}{u^H u} = u u^H.$$

- the component of v orthogonal to u equals

$$v - \frac{u^H v}{u^H u} u = v - u^H v u.$$

- the matrix that projects a vector onto the space orthogonal to u is given by

$$I - \frac{u u^H}{u^H u} = I - u u^H.$$

Homework 2.2.3.1. Let $u \neq 0 \in \mathbb{C}^m$.

ALWAYS/SOMETIMES/NEVER $u/\|u\|_2$ has unit length.

▼ Answer ▼ Solution

$$\left\| \frac{u}{\|u\|_2} \right\|_2$$

$= \quad < \text{ homogeneity of norms } >$

$$\frac{\|u\|_2}{\|u\|_2}$$

$= \quad < \text{ algebra } >$

$$1$$

ALWAYS.

Now prove it.

This last exercise shows that any nonzero vector can be scaled (normalized) to have unit length.

Definition 2.2.3.1. Orthonormal vectors. Let $u_0, u_1, \ldots, u_{n-1} \in \mathbb{C}^m$. These vectors are said to be mutually orthonormal if for all $0 \leq i, j < n$

$$u_i^H u_j = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{otherwise} \end{cases}.$$

The definition implies that $\|u_i\|_2 = \sqrt{u_i^H u_i} = 1$ and hence each of the vectors is of unit length in addition to being orthogonal to each other.

The standard basis vectors (Definition 1.3.1.3)

$$\{e_j\}_{i=0}^{m-1} \subset \mathbb{C}^m,$$

where

$$e_j = \begin{pmatrix} 0 \\ \vdots \\ 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix} \longleftarrow \text{ entry indexed with } j$$

are mutually orthonormal since, clearly,

$$e_i^H e_j = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{otherwise.} \end{cases}$$

Naturally, any subset of the standard basis vectors is a set of mutually orthonormal vectors.

> **Remark 2.2.3.2.** For n vectors of size m to be mutually orthonormal, n must be less than or equal to m. This is because n mutually orthonormal vectors are linearly independent and there can be at most m linearly independent vectors of size m.

A very concise way of indicating that a set of vectors are mutually orthonormal is to view them as the columns of a matrix, which then has a very special property:

> **Definition 2.2.3.3. Orthonormal matrix.** Let $Q \in \mathbb{C}^{m \times n}$ (with $n \leq m$). Then Q is said to be an orthonormal matrix iff $Q^H Q = I$.

The subsequent exercise makes the connection between mutually orthonormal vectors and an orthonormal matrix.

Homework 2.2.3.2. Let $Q \in \mathbb{C}^{m \times n}$ (with $n \leq m$). Partition $Q = \left(\begin{array}{c|c|c|c} q_0 & q_1 & \cdots & q_{n-1} \end{array} \right)$.

TRUE/FALSE: Q is an orthonormal matrix if and only if $q_0, q_1, \ldots, q_{n-1}$ are mutually orthonormal.

▼ Answer ▼ Solution

Let $Q \in \mathbb{C}^{m \times n}$ (with $n \leq m$). Partition $Q = \left(\begin{array}{c|c|c|c} q_0 & q_1 & \cdots & q_{n-1} \end{array} \right)$. Then

$$\begin{array}{rcl} Q^H Q & = & \left(\begin{array}{c|c|c|c} q_0 & q_1 & \cdots & q_{n-1} \end{array} \right)^H \left(\begin{array}{c|c|c|c} q_0 & q_1 & \cdots & q_{n-1} \end{array} \right) \\ & = & \left(\begin{array}{c} q_0^H \\ q_1^H \\ \vdots \\ q_{n-1}^H \end{array} \right) \left(\begin{array}{c|c|c|c} q_0 & q_1 & \cdots & q_{n-1} \end{array} \right) \\ & = & \left(\begin{array}{c|c|c|c} q_0^H q_0 & q_0^H q_1 & \cdots & q_0^H q_{n-1} \\ \hline q_1^H q_0 & q_1^H q_1 & \cdots & q_1^H q_{n-1} \\ \hline \vdots & \vdots & & \vdots \\ \hline q_{n-1}^H q_0 & q_{n-1}^H q_1 & \cdots & q_{n-1}^H q_{n-1} \end{array} \right). \end{array}$$

Now consider that $Q^H Q = I$:

$$\left(\begin{array}{c|c|c|c} q_0^H q_0 & q_0^H q_1 & \cdots & q_0^H q_{n-1} \\ \hline q_1^H q_0 & q_1^H q_1 & \cdots & q_1^H q_{n-1} \\ \hline \vdots & \vdots & & \vdots \\ \hline q_{n-1}^H q_0 & q_{n-1}^H q_1 & \cdots & q_{n-1}^H q_{n-1} \end{array}\right) = \left(\begin{array}{c|c|c|c} 1 & 0 & \cdots & 0 \\ \hline 0 & 1 & \cdots & 0 \\ \hline \vdots & \vdots & & \vdots \\ \hline 0 & 0 & \cdots & 1 \end{array}\right).$$

Clearly Q is orthonormal if and only if $q_0, q_1, \ldots, q_{n-1}$ are mutually orthonormal.

TRUE

Now prove it!

Homework 2.2.3.3. Let $Q \in \mathbb{C}^{m \times n}$.

ALWAYS/SOMETIMES/NEVER: If $Q^H Q = I$ then $QQ^H = I$.

▼ Answer ▼ Solution

- If Q is a square matrix ($m = n$) then $Q^H Q = I$ means $Q^{-1} = Q^H$. But then $QQ^{-1} = I$ and hence $QQ^H = I$.
- If Q is not square, then $Q^H Q = I$ means $m > n$. Hence Q has rank equal to n which in turn means QQ^H is a matrix with rank at most equal to n. (Actually, its rank equals n.). Since I has rank equal to m (it is an $m \times m$ matrix with linearly independent columns), QQ^H cannot equal I.

  More concretely: let $m > 1$ and $n = 1$. Choose $Q = \left(e_0 \right)$. Then $Q^H Q = e_0^H e_0 = 1 = I$. But

$$QQ^H = e_0 e_0^H = \begin{pmatrix} 1 & 0 & \cdots \\ 0 & 0 & \cdots \\ \vdots & \vdots & \end{pmatrix}.$$

SOMETIMES.

Now explain why.